

WASP-39b

R-1603

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-39_160520 · created 2026-08-02T02:37:26+00:00

PRACTICE / VALIDATION RUN — not submitted to any science database

Results

Mid-Transit Time (Tmid)	2457528.714 +/- 0.04 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.19 +/- 0.053
Transit depth (Rp/Rs)^2	3.62 +/- 2.03 [%]
Orbital Inclination (inc)	89.8 +/- 2.13
Airmass coefficient 1 (a1)	1.35 +/- 0.68
Airmass coefficient 2 (a2)	-0.31 +/- 0.45
Scatter in the residuals of the lightcurve fit is	41.92 %
Best Comparison Star	None
Optimal Aperture	7.75
Optimal Annulus	31.83
Transit Duration (day)	0.064 +/- 0.036

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.19 ± 0.053 vs 0.1457 (deviation 0.84σ)
Duration fit vs published	0.064 d vs 0.12 d (deviation 1.56σ)
Mid-time O–C	-74.4 min
Depth SNR	3.6
Residual scatter	41.92 %

Light curve

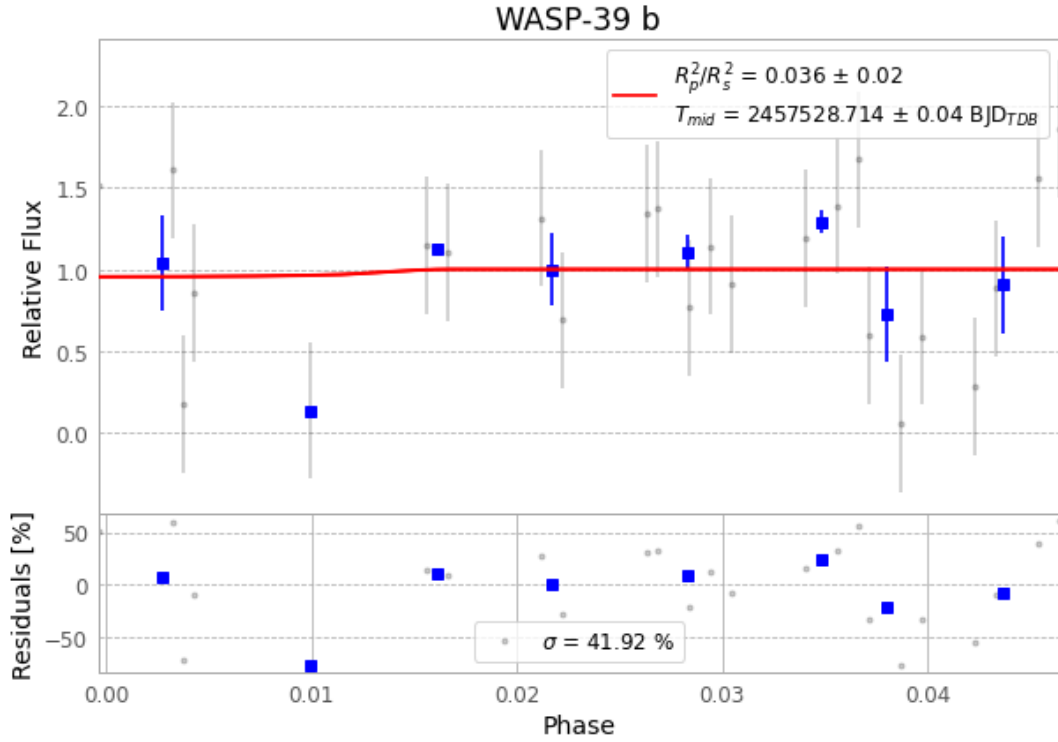


Figure 1 — FinalLightCurve_WASP-39 b_2016-05-19.png (SHA-256 015c76cea3bd44df...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [278, 42], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 912ab2c16533dc4a...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

98 FITS frames (64 MB), WASP-39160520043912.FITS → WASP-39160520093012.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-39-160520.log (f80369d43675...), FinalLightCurve_WASP-39 b_2016-05-19.png (015c76cea3bd...), FinalLightCurve_WASP-39 b_2016-05-19.csv (73b30d0e01c3...)

Compute resources

Wall time 245.5 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 02:37Z.